Al-Balqa Applied University
Computer Engineering Department

Faculty of Engineering Technology
Digital logic design – Midterm Exam

Date: Monday 20, 11/2017

Name:

Q1) choose the correct answer (12 marks)

1	2	3	4	5	6	7	8

1. Which of the following number equivalent to $(276)_O$?
   a. $(276)_H$ b) $(BE)_H$ c) 010111101 d) $(192)_6$
2. Which of the following number equivalent to $(1ABF)_H$?
   a) 1101010111111 b) 1010101111111 c) 1101010111111 d) 1101010011111
3. The maximum value can be represented from 9 bit number is
   a) 511 b) 512 c) 1024 d) 1023
4. Which of the following function is equivalent to $F(A,B,C) = \sum_m (1,3,4,5)$
   A. $F(A,B,C) = A'B'C' + A'BC + AB'C' + ABC$
   B. $F(A,B,C) = A'B'C + A'BC + AB'C' + AB'C$
   C. $F(A,B,C) = A'B'C' + AB'C + A'BC + ABC$
   D. $F(A,B,C) = A'B'C' + AB'C + A'BC' + ABC'$

5. Which of the following number is equivalent to (328)10
   a) 101001000 b) 101001100 c) 101011100 d) 101101000

6. The 10th complement of 2643 number is
   a) 7356 b) 7357 c) 7355 d) 7358
7. Which of the following function will be get from the k-map
   a) $F = \sum(2,5,11,12)$ b) $F = \sum(2,5,8,15)$ c) $F = \sum(3,5,10,12)$
   d) $F = \sum(3,5,8,14)$

AB \ CD	00	01	11	10
00				1
01		1		
11	1			
10			1	

8. We cant get the full adder from
   a) 2 half adder and 2 OR gate
   b) one half adder and 2 AND gate
   c) 2 half adder and 1 OR gate
   D) 2 half adder and 1 AND gate

Q2) fill in gap (6 marks)

1. Convert decimal 59.72 to BCD ....
2. To implement XNOR gate using minimum number NANDs. How many NANDs we need?
   ....
3. An equivalent representation for Boolean AB+1 is:
4. An equivalent representation of $F(A,B,C)=(AB+C).C'$ is:
5. Convert $(1100101000110101)_2$ to hexadecimal ....
6. If $F(A,B,C,D)=\sum_m(0,1,8,9,11)$; the $F' = \prod M($ $)$

Q3) what is the simplified function of the following K-map using product of sum (4 marks)

AB \ CD	00	01	11	10
00	1	1		X
01	x	x		
11				
10				1

Q4) Design the circuit needed to implement the following where x, y and z are 3bits numbers ,using block adder, comparator and basic gate? 8 marks

If(x>=2)

F=(x+y)*z;

Else

F=(x-y)*z;

Q5) **Find $(-112)_{10} - (78)_{10}$: 4marks**

a. Using 1's compliment

b. Using 2's compliment

**Q6) Design a combinational circuit that takes a 3 bit number $X=X_3X_2X_1$, and one output (
F=1 when the input is dividable by 3 with remainder =0 *OR* when the XNOR of the 3
number is 1 ; otherwise F=0. (6 marks)**

Team
ZerOne
Computer & Networks Engineers

Q1:-

① $(276)_{octal} \Rightarrow (\underline{010}\ \underline{111}\ \underline{110}) \Rightarrow (\overset{1011}{\overset{\uparrow}{B}}\ \overset{1110}{\overset{\uparrow}{E}})_{16}$

كل رقم ٣ بت   كل ٤ بت رقم بال hexa-deci

(b)

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② $(1ABF)_H \Rightarrow (0001\ 1010\ 1011\ 1111)_2$

كل رقم ٤ بت

(c)

(15) F → 1111
(11) B → 1011
(10) A → 1010
1 → 0001

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③ $2^9 - 1 = 511$ (a)

rule

$\boxed{2^{\text{number of bit}} - 1}$

وممكن بالتجريب

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④ $F(A,B,C) = \sum m\,(1,3,4,5) = \bar{A}\bar{B}C + \bar{A}BC + A\bar{B}\bar{C} + A\bar{B}C$

(b)

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⑤ $(328)_{10} \Rightarrow (101001000)$ (a)

أو قسمة متكررة

256	128	64	32	16	8	4	2	1
1	0	1	0	0	1	0	0	0

1

⑥ 2643

9's comp → 7356

$$\begin{array}{r} 7356 \\ +\ 1 \\ \hline 7357 \end{array}$$

10^{th} comp → 7357 ⓐ

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⑦

AB \ CD	00	01	11	10
00				1
01		1		
11	1			
10			1	

AB CD
0010 → (2)

5 ← 0101

12 ← 1100

1011 → (11)

$F = \Sigma(2, 5, 11, 12)$ ⓐ

---

⑧ we can get the full adder from :- (عدلوا بالإمتحان)

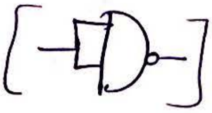
- 2 half adder and 1 OR gate.

Q2 :- ① $(59.72)_{10} \Rightarrow (0101\ 1001.0111\ 0010)_{BCD}$

كل رقم ٤ بت بال BCD

② XNOR by Nand Gates (أقل عدد) ?!

* أقل عدد كائن لتمثيل XOR من خلال ال Nand هو ٤ حتى أمثل XNOR

بس بنفي المخرج من خلال ال Nand [] ← وهاي الخامسة ،

وبالتالي أقل عدد هو 5

③ $AB + 1 = 1$

④ $(AB + C) \cdot \overline{C} \Rightarrow AB\overline{C} + \underbrace{\cancel{C\overline{C}}}_{=0} = AB\overline{C}$

بوزع الضرب على الجمع

⑤ $(\underline{1100}\,\underline{1010}\,\underline{0011}\,\underline{0101})_2 \Rightarrow (CA35)_{16}$

$$\begin{array}{cccc} 1100 & 1010 & 0011 & 0101 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ C & A & 3 & 5 \end{array}$$

⑥ $F(A,B,C,D) = \sum_m (0,1,8,9,11)$

المطلوب $\boxed{\overline{F}} = \prod M(0,1,8,9,11)$ ← نفهم

← توقّع ...

	A	B	F	$\bar{F}$
m_0	0	0	1	0
	0	1	0	
m_2	1	0	1	0
	1	1	0	

<< ع اشي أبسط

max term $\prod M$

لهم نفسهم ال minterm وبين قاعدة عموماً.

Q3:-

AB \ CD	00	01	11	10
00	1	1	0	X
01	X=0	X=0	0	0
11	0	0	0	0
10	0	0	0	1

Group 1, Group 2, Group 3

POS ?!

0 → بنفسه

1 → comp

$$F = (\bar{C}+\bar{D}) \cdot (\bar{B}) \cdot (C+\bar{A})$$

G1 G2 G3

* أو ممكن أحل SOP وأحولها لـ POS.

Q4:

(أكبر أو يساوي)

نعيد ترتيب الأفكار

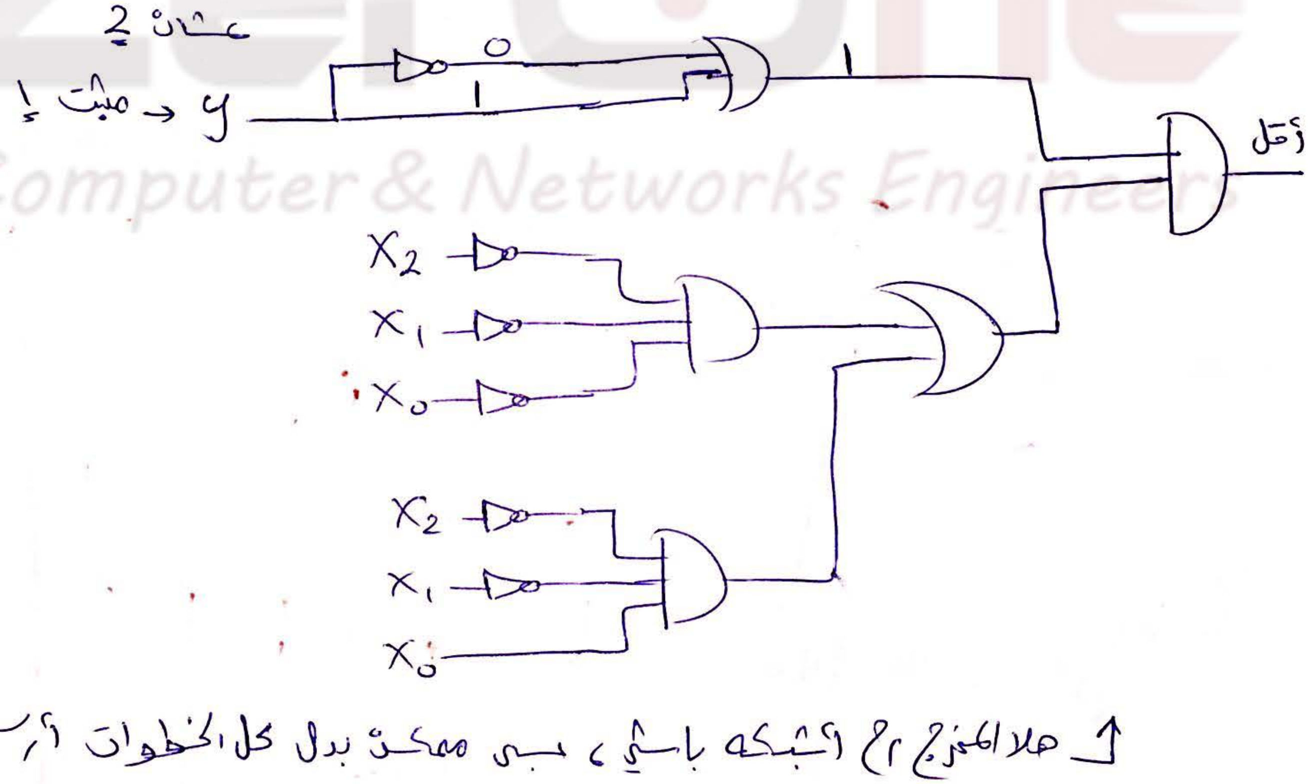
* لو كان أقل بعمل $(x-y)*z$ و else بعمل $(x+y)*z$

$\boxed{X \rightarrow 3bit}$

x_2	x_1	x_0	$\underline{2}$ 1	0	أقل <
0	0	0	1	1	1
0	0	1	1	1	1
0	1	0	1	1	0

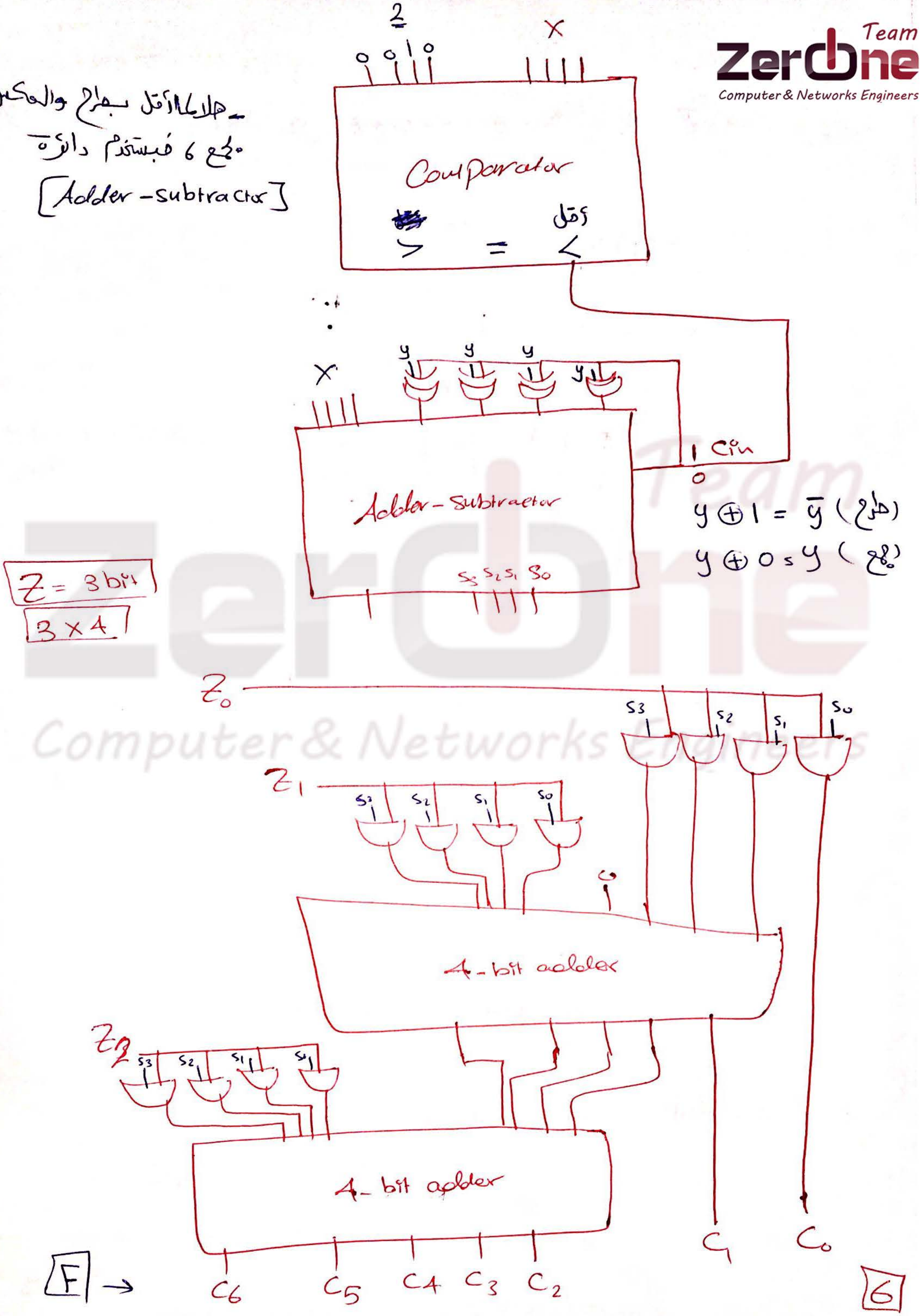
* يعني أقل من 2!
بس لحالتين لما يكون zero و one

$$\boxed{F = \bar{x}_2\bar{x}_1\bar{x}_0 + \bar{x}_2\bar{x}_1x_0}$$



↑ هذا المخرج ع الشبكة ماشي، بس ممكن بدل كل الخطوات أرسم
IC الـ Comparator مباشر >>

ـ هل الأقل نطرح والعكس
نجمع فنستخدم دائرة
[Adder-subtractor]



Q5 :-

$(-112)_{10} - (78)_{10} = \ !?$ المفروض -190

@

$(112)_{10} \Rightarrow (01110000)_2 \Longrightarrow$ first comp $(1000\ 1111)$

زدت bit عشان مايصير over flow

64	32	16	8	4	2	1
1	1	1	0	0	0	0

$(78)_{10} \Rightarrow (01001110)_2 \Longrightarrow$ first comp $(1011\ 0001)$

زدت bit عشان مايصير over flow

64	32	16	8	4	2	1
1	0	0	1	1	1	0

$\boxed{-112}$ 1000 1111

$\boxed{-78}$ 1011 0001 +

1 01000000

1 +

01000001

الحل غلط

لو أخذت 1st comp لكن أتأكد

128 64 32 16 8 4 2 1

10111110 => 190

(b)

$(112)_{10} \rightarrow (0111\ 0000)_2 \xrightarrow{\text{1st comp}} (1000\ 1111)$

1 +

$\boxed{(10010000)}$

$(78)_{10} \rightarrow (01001110)_2 \xrightarrow{\text{1st comp}} (1011\ 0001)$

1 +

$\boxed{(10110010)}$

$\boxed{-112}$ 1001 0000

$\boxed{-78}$ 1011 0010 +

X 01000010 ⟶ 10111101

ل سالب

لو بدي أتأكد باخد 1st comp وبزيد واحد

10111101
1+

10111110

⇓

190 ✓

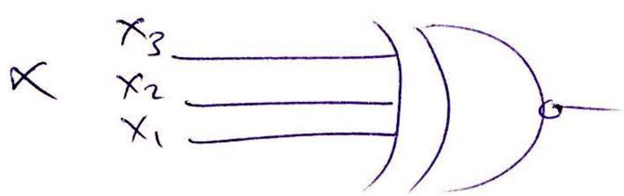
Q6 :-

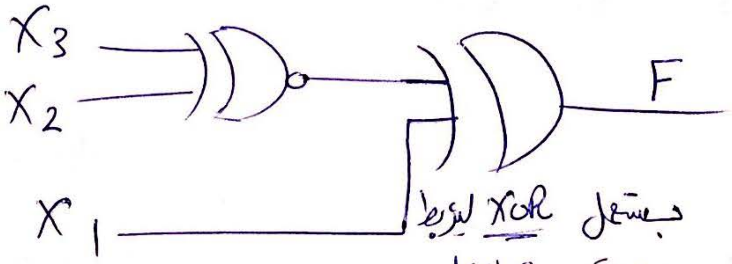
$\boxed{F}$

X_3	X_2	X_1	dividable by 3	XNOR	/3 oR XNOR
0	0	0	1	1	1
0	0	1	0	0	0
0	1	0	0	0	0
0	1	1	1	1	1
1	0	0	0	0	0
1	0	1	0	1	1
1	1	0	1	1	1
1	1	1	0	0	0

نفسه

لكن أنتبه انه XNOR مالها لامداخل ومش تجميعة ...





بستعمل XOR لربط ومكن تجرب من خلال انه تكافؤ ToT